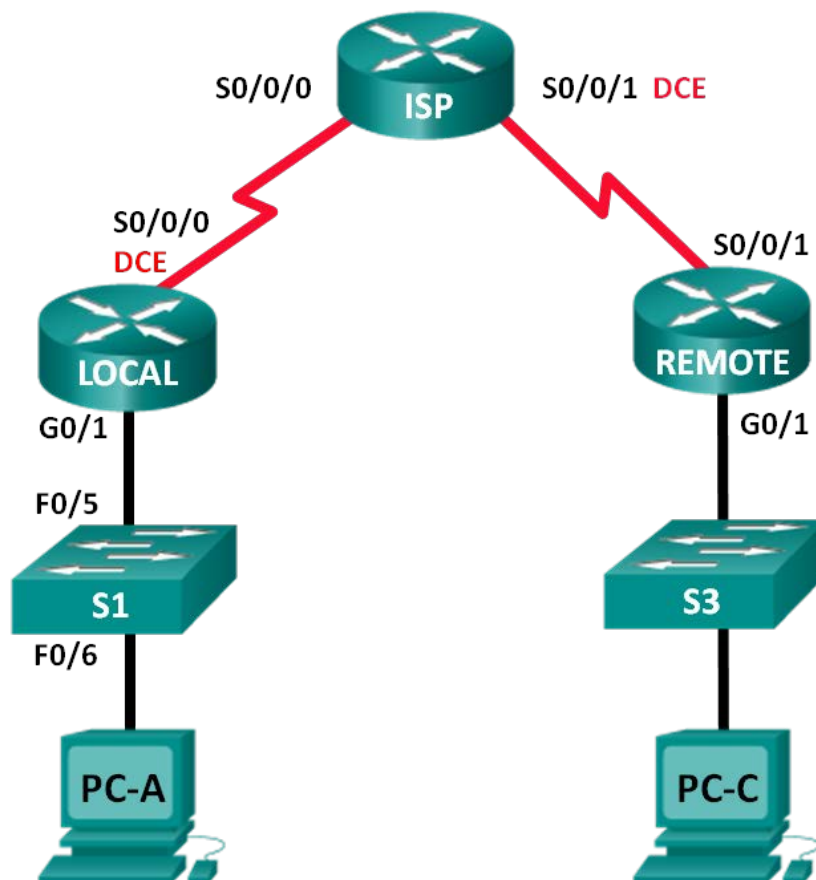


Lab – Testing Network Connectivity with Ping and Traceroute (Instructor Version – Optional Lab)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only. Optional activities are designed to enhance understanding and/or to provide additional practice.

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
LOCAL	G0/1	192.168.1.1	255.255.255.0	N/A
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A
ISP	S0/0/0	10.1.1.2	255.255.255.252	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A
REMOTE	G0/1	192.168.3.1	255.255.255.0	N/A
	S0/0/1	10.2.2.1	255.255.255.252	N/A
S1	VLAN 1	192.168.1.11	255.255.255.0	192.168.1.1
S3	VLAN 1	192.168.3.11	255.255.255.0	192.168.3.1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1

Objectives

Part 1: Build and Configure the Network

Part 2: Use Ping Command for Basic Network Testing

Part 3: Use Tracert and Traceroute Commands for Basic Network Testing

Part 4: Troubleshoot the Topology

Background / Scenario

Ping and traceroute are two tools that are indispensable when testing TCP/IP network connectivity. Ping is a network administration utility used to test the reachability of a device on an IP network. This utility also measures the round-trip time for messages sent from the originating host to a destination computer. The ping utility is available on Windows, Unix-like operating systems (OS), and the Cisco Internetwork Operating System (IOS).

The traceroute utility is a network diagnostic tool for displaying the route and measuring the transit delays of packets travelling an IP network. The tracert utility is available on Windows, and a similar utility, traceroute, is available on Unix-like OS and Cisco IOS.

In this lab, the **ping** and **traceroute** commands are examined and command options are explored to modify the command behavior. Cisco devices and PCs are used in this lab for command exploration. Cisco routers will use Enhanced Interior Gateway Routing Protocol (EIGRP) to route packets between networks. The necessary Cisco device configurations are provided in this lab.

Note: The routers used with CCNA hands-on labs are Cisco 1941 Integrated Services Routers (ISRs) with Cisco IOS Release 15.2(4)M3 (universalk9 image). The switches used are Cisco Catalyst 2960s with Cisco IOS Release 15.0(2) (lanbasek9 image). Other routers, switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of this lab for the correct interface identifiers.

Note: Make sure that the routers and switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Instructor Note: Refer to the Instructor Lab Manual for the procedures to initialize and reload devices.

Required Resources

- 3 Routers (Cisco 1941 with Cisco IOS Release 15.2(4)M3 universal image or comparable)
- 2 Switches (Cisco 2960 with Cisco IOS Release 15.0(2) lanbasek9 image or comparable)
- 2 PCs (Windows 7 or 8 with terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

Part 1: Build and Configure the Network

In Part 1, you will set up the network in the topology and configure the PCs and Cisco devices. The initial configurations for the routers and switches are provided for your reference. In this topology, EIGRP is used to route packets between networks.

Step 1: Cable the network as shown in the topology.

Step 2: Erase the configurations on the routers and switches, and reload the devices.

Step 3: Configure PC IP addresses and default gateways according to the Addressing Table.

Step 4: Configure the LOCAL, ISP, and REMOTE routers using the initial configurations provided below.

At the switch or router global config mode prompt, copy and paste the configuration for each device. Save the configuration to startup-config.

Instructor Note: The command “no auto-summary” for EIGRP is included for compatibility with older routers and IOS versions. With the 1941 router and IOS 15 specified for this lab, no auto-summary is the default,

Initial configurations for the LOCAL router:

```
hostname LOCAL
no ip domain-lookup
interface s0/0/0
 ip address 10.1.1.1 255.255.255.252
 clock rate 56000
 no shutdown
interface g0/1
 ip add 192.168.1.1 255.255.255.0
 no shutdown
router eigrp 1
 network 10.1.1.0 0.0.0.3
 network 192.168.1.0 0.0.0.255
 no auto-summary
```

Initial configurations for ISP:

```
hostname ISP
no ip domain-lookup
interface s0/0/0
 ip address 10.1.1.2 255.255.255.252
```

```
no shutdown
interface s0/0/1
ip add 10.2.2.2 255.255.255.252
clock rate 56000
no shutdown
router eigrp 1
network 10.1.1.0 0.0.0.3
network 10.2.2.0 0.0.0.3
no auto-summary
end
```

Initial configurations for REMOTE:

```
hostname REMOTE
no ip domain-lookup
interface s0/0/1
ip address 10.2.2.1 255.255.255.252
no shutdown
interface g0/1
ip add 192.168.3.1 255.255.255.0
no shutdown
router eigrp 1
network 10.2.2.0 0.0.0.3
network 192.168.3.0 0.0.0.255
no auto-summary
end
```

Step 5: Configure the S1 and S3 switches with the initial configurations.

Instructor Note: If Netlab is used, switch interface FastEthernet 0/1 – 0/4 should be shutdown for this lab. Use the following commands on S1 and S3:

```
Switch (config)# interface range f0/1 - 4
Switch (config)# shutdown
```

Initial configurations for S1:

```
hostname S1
no ip domain-lookup
interface vlan 1
ip add 192.168.1.11 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.1.1
end
```

Initial configurations for S3:

```
hostname S3
no ip domain-lookup
interface vlan 1
ip add 192.168.3.11 255.255.255.0
```

```
no shutdown
exit
ip default-gateway 192.168.3.1
end
```

Step 6: Configure an IP host table on the LOCAL router.

The IP host table allows you to use a hostname to connect to a remote device rather than an IP address. The host table provides name resolution for the device with the following configurations. Copy and paste the following configurations for the LOCAL router. The configurations will allow you to use the hostnames for **ping** and **traceroute** commands on the LOCAL router.

```
ip host REMOTE 10.2.2.1 192.168.3.1
ip host ISP 10.1.1.2 10.2.2.2
ip host LOCAL 192.168.1.1 10.1.1.1
ip host PC-C 192.168.3.3
ip host PC-A 192.168.1.3
ip host S1 192.168.1.11
ip host S3 192.168.3.11
end
```

Part 2: Use Ping Command for Basic Network Testing

In Part 2 of this lab, use the **ping** command to verify end-to-end connectivity. Ping operates by sending Internet Control Message Protocol (ICMP) echo request packets to the target host and then waiting for an ICMP response. It can record the round trip time and any packet loss.

You will examine the results with the **ping** command and the additional ping options that are available on Windows-based PCs and Cisco devices.

Step 1: Test network connectivity from the LOCAL network using PC-A.

All the pings from PC-A to other devices in the topology should be successful. If they are not, check the topology and the cabling, as well as the configuration of the Cisco devices and the PCs.

- a. Ping from PC-A to its default gateway (LOCAL's GigabitEthernet 0/1 interface).

```
C:\Users\User1> ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 192.168.1.1:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

In this example, four (4) ICMP requests, 32 bytes each, were sent and the responses were received in less than one millisecond with no packet loss. The transmission and reply time increases as the ICMP requests and responses are processed by more devices during the journey to and from the final destination.

- b. From PC-A, ping the addresses listed in the following table and record the average round trip time and Time to Live (TTL).

Destination	Average Round Trip Time (ms)	TTL
192.168.1.1 (LOCAL)	0	255
192.168.1.11 (S1)	0*	255
10.1.1.1 (LOCAL)	0	255
10.1.1.2 (ISP)	20	254
10.2.2.2 (ISP)	20	254
10.2.2.1 (REMOTE)	40	253
192.168.3.1 (REMOTE)	40	253
192.168.3.11 (S3)	40*	252
192.168.3.3 (PC-C)	40	125

***Instructor Note:** The average round trip time was increased if the message “Request timed out” was displayed during the first ICMP request. The delay was caused by ARP, and this resulted in packet loss.

Notice the average round trip time to 192.168.3.3 (PC-C). The time increased because the ICMP requests were processed by three routers before PC-A received the reply from PC-C.

```
C:\Users\User1> ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125
Reply from 192.168.3.3: bytes=32 time=40ms TTL=125
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125
```

```
Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 41ms, Average = 40ms
```

Step 2: Use extended ping commands on a PC.

The default **ping** command sends four requests at 32 bytes each. It waits 4,000 milliseconds (4 seconds) for each response to be returned before displaying the “Request timed out” message. The **ping** command can be fine tuned for troubleshooting a network.

- a. At the command prompt, type **ping** and press Enter.

```
C:\Users\User1> ping
Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
          [-r count] [-s count] [[-j host-list] | [-k host-list]]
          [-w timeout] [-R] [-S srcaddr] [-4] [-6] target_name
```

Options:

```
-t           Ping the specified host until stopped.
            To see statistics and continue - type Control-Break;
            To stop - type Control-C.
```

-a	Resolve addresses to hostnames.
-n count	Number of echo requests to send.
-l size	Send buffer size.
-f	Set Don't Fragment flag in packet (IPv4-only).
-i TTL	Time To Live.
-v TOS	Type Of Service (IPv4-only. This setting has been deprecated and has no effect on the type of service field in the IP Header).
-r count	Record route for count hops (IPv4-only).
-s count	Timestamp for count hops (IPv4-only).
-j host-list	Loose source route along host-list (IPv4-only).
-k host-list	Strict source route along host-list (IPv4-only).
-w timeout	Timeout in milliseconds to wait for each reply.
-R	Use routing header to test reverse route also (IPv6-only).
-S srcaddr	Source address to use.
-4	Force using IPv4.
-6	Force using IPv6.

- b. Using the **-t** option, ping PC-C to verify that PC-C is reachable.

```
C:\Users\User1> ping -t 192.168.3.3
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125
Reply from 192.168.3.3: bytes=32 time=40ms TTL=125
```

To illustrate the results when a host is unreachable, disconnect the cable between the REMOTE router and the S3 switch, or shut down the GigabitEthernet 0/1 interface on the REMOTE router.

```
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125
Reply from 192.168.1.3: Destination host unreachable.
Reply from 192.168.1.3: Destination host unreachable.
```

While the network is functioning correctly, the **ping** command can determine whether the destination responded and how long it took to receive a reply from the destination. If a network connectivity problem exists, the **ping** command displays an error message.

- c. Reconnect the Ethernet cable or enable the GigabitEthernet interface on the REMOTE router (using the **no shutdown** command) before moving onto the next step. After about 30 seconds, the ping should be successful again.

```
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125
Reply from 192.168.3.3: bytes=32 time=40ms TTL=125
```

- d. Press **Ctrl+C** to stop the ping command.

Step 3: Test network connectivity from the LOCAL network using Cisco devices.

The **ping** command is also available on Cisco devices. In this step, the **ping** command is examined using the LOCAL router and the S1 switch.

- a. Ping PC-C on the REMOTE network using the IP address of 192.168.3.3 from the LOCAL router.

```
LOCAL# ping 192.168.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/64/68 ms
```

The exclamation point (!) indicates that the ping was successful from the LOCAL router to PC-C. The round trip takes an average of 64 ms with no packet loss, as indicated by a 100% success rate.

- b. Because a local host table was configured on the LOCAL router, you can ping PC-C on the REMOTE network using the hostname configured from the LOCAL router.

```
LOCAL# ping PC-C
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/63/64 ms
```

- c. There are more options available for the **ping** command. At the CLI, type **ping** and press Enter. Input **192.168.3.3** or **PC-C** for the Target IP address. Press Enter to accept the default value for other options.

```
LOCAL# ping
```

```
Protocol [ip]:
```

```
Target IP address: PC-C
```

```
Repeat count [5]:
```

```
Datagram size [100]:
```

```
Timeout in seconds [2]:
```

```
Extended commands [n]:
```

```
Sweep range of sizes [n]:
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/63/64 ms
```

- d. You can use an extended ping to observe when there is a network issue. Start the **ping** command to 192.168.3.3 with a repeat a count of 500. Then, disconnect the cable between the REMOTE router and the S3 switch or shut down the GigabitEthernet 0/1 interface on the REMOTE router.

Reconnect the Ethernet cable or enable the GigabitEthernet interface on the REMOTE router after the exclamation points (!) have replaced by the letter U and periods (.). After about 30 seconds, the ping should be successful again. Press **Ctrl+Shift+6** to stop the **ping** command if desired.

```
LOCAL# ping
```

```
Protocol [ip]:
```

```
Target IP address: 192.168.3.3
```

```
Repeat count [5]: 500
```

```
Datagram size [100]:
```

```
Timeout in seconds [2]:
```

```
Extended commands [n]:
```

```
Sweep range of sizes [n]:
```

```
Type escape sequence to abort.
```

```
Sending 500, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds:
```

```
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!U.....
...!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```



```
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!
```

Success rate is 95 percent (479/500), round-trip min/avg/max = 60/63/72 ms

The letter U in the results indicates that a destination is unreachable. An error protocol data unit (PDU) was received by the LOCAL router. Each period (.) in the output indicates that the ping timed out while waiting for a reply from PC-C. In this example, 5% of the packets were lost during the simulated network outage.

Note: You can also use the following command for the same results:

```
LOCAL# ping 192.168.3.3 repeat 500
```

or

```
LOCAL# ping PC-C repeat 500
```

- e. You can also test network connectivity with a switch. In this example, the S1 switch pings the S3 switch on the REMOTE network.

```
S1# ping 192.168.3.11
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.11, timeout is 2 seconds:

```
!!!!!!
```

Success rate is 100 percent (5/5), round-trip min/avg/max = 67/67/68 ms

The **ping** command is extremely useful when troubleshooting network connectivity. However, ping cannot indicate the location of problem when a ping is not successful. The **tracert** (or **traceroute**) command can display network latency and path information.

Part 3: Use Tracert and Traceroute Commands for Basic Network Testing

The commands for tracing routes can be found on PCs and network devices. For a Windows-based PC, the **tracert** command uses ICMP messages to trace the path to the final destination. The **traceroute** command utilizes the User Datagram Protocol (UDP) datagrams for tracing routes to the final destination for Cisco devices and other Unix-like PCs.

In Part 3, you will examine the traceroute commands and determine the path that a packet travels to its final destination. You will use the **tracert** command from the Windows PCs and the **traceroute** command from the Cisco devices. You will also examine the options that are available for fine tuning the traceroute results.

Step 1: Use the tracert command from PC-A to PC-C.

- a. At the command prompt, type **tracert 192.168.3.3**.

```
C:\Users\User1> tracert 192.168.3.3
```

Tracing route to PC-C [192.168.3.3]

Over a maximum of 30 hops:

1	<1 ms	<1 ms	<1 ms	192.168.1.1
2	24 ms	24 ms	24 ms	10.1.1.2
3	48 ms	48 ms	48 ms	10.2.2.1
4	59 ms	59 ms	59 ms	PC-C [192.168.3.3]

Trace complete.

The tracert results indicates the path from PC-A to PC-C is from PC-A to LOCAL to ISP to REMOTE to PC-C. The path to PC-C traveled through three router hops to the final destination of PC-C.

Step 2: Explore additional options for the tracert command.

- a. At the command prompt, type **tracert** and press Enter.

```
C:\Users\User1> tracert
```

```
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
          [-R] [-S srcaddr] [-4] [-6] target_name
```

Options:

-d	Do not resolve addresses to hostnames.
-h maximum_hops	Maximum number of hops to search for target.
-j host-list	Loose source route along host-list (IPv4-only).
-w timeout	Wait timeout milliseconds for each reply.
-R	Trace round-trip path (IPv6-only).
-S srcaddr	Source address to use (IPv6-only).
-4	Force using IPv4.
-6	Force using IPv6.

- b. Use the **-d** option. Notice that the IP address of 192.168.3.3 is not resolved as PC-C.

```
C:\Users\User1> tracert -d 192.168.3.3
```

```
Tracing route to 192.168.3.3 over a maximum of 30 hops:
```

1	<1 ms	<1 ms	<1 ms	192.168.1.1
2	24 ms	24 ms	24 ms	10.1.1.2
3	48 ms	48 ms	48 ms	10.2.2.1
4	59 ms	59 ms	59 ms	192.168.3.3

```
Trace complete.
```

Step 3: Use the traceroute command from the LOCAL router to PC-C.

- a. At the command prompt, type **traceroute 192.168.3.3** or **traceroute PC-C** on the LOCAL router. The hostnames are resolved because a local IP host table was configured on the LOCAL router.

```
LOCAL# traceroute 192.168.3.3
```

```
Type escape sequence to abort.
```

```
Tracing the route to PC-C (192.168.3.3)
```

```
VRF info: (vrf in name/id, vrf out name/id)
```

1	ISP (10.1.1.2)	16 msec	16 msec	16 msec
2	REMOTE (10.2.2.1)	28 msec	32 msec	28 msec
3	PC-C (192.168.3.3)	32 msec	28 msec	32 msec

```
LOCAL# traceroute PC-C
```

```
Type escape sequence to abort.
```

```
Tracing the route to PC-C (192.168.3.3)
```

```
VRF info: (vrf in name/id, vrf out name/id)
```

1	ISP (10.1.1.2)	16 msec	16 msec	16 msec
2	REMOTE (10.2.2.1)	28 msec	32 msec	28 msec
3	PC-C (192.168.3.3)	32 msec	32 msec	28 msec

Step 4: Use the traceroute command from the S1 switch to PC-C.

- a. On the S1 switch, type **traceroute 192.168.3.3**. The hostnames are not displayed in the traceroute results because a local IP host table was not configured on this switch.

```
S1# traceroute 192.168.3.3
Type escape sequence to abort.
Tracing the route to 192.168.3.3
VRF info: (vrf in name/id, vrf out name/id)
 1 192.168.1.1 1007 msec 0 msec 0 msec
 2 10.1.1.2 17 msec 17 msec 16 msec
 3 10.2.2.1 34 msec 33 msec 26 msec
 4 192.168.3.3 33 msec 34 msec 33 msec
```

The **traceroute** command has additional options. You can use the **?** or just press Enter after typing **traceroute** at the prompt to explore these options.

The following link provides more information regarding the **ping** and **traceroute** commands for a Cisco device:

http://www.cisco.com/en/US/products/sw/iosswrel/ps1831/products_tech_note09186a00800a6057.shtml

Part 4: Troubleshoot the Topology

Step 1: Erase the configurations on the REMOTE router.

Step 2: Reload the REMOTE router.

Step 3: Copy and paste the following configuration into the REMOTE router.

```
hostname REMOTE
no ip domain-lookup
interface s0/0/1
 ip address 10.2.2.1 255.255.255.252
 no shutdown
interface g0/1
 ip add 192.168.8.1 255.255.255.0
 no shutdown
router eigrp 1
 network 10.2.2.0 0.0.0.3
 network 192.168.3.0 0.0.0.255
 no auto-summary
end
```

Step 4: From the LOCAL network, use ping and tracert or traceroute commands to troubleshoot and correct the problem on the REMOTE network.

- a. Use the **ping** and **tracert** commands from PC-A.

You can use the **tracert** command to determine end-to-end network connectivity. This tracert result indicates that PC-A can reach its default gateway of 192.168.1.1, but PC-A does not have network connectivity with PC-C.

```
C:\Users\User1> tracert 192.168.3.3
```

Tracing route to 192.168.3.3 over a maximum of 30 hops

```
1    <1 ms    <1 ms    <1 ms    192.168.1.1
2    192.168.1.1  reports: Destination host unreachable.
```

Trace complete.

One way to locate the network issue is to ping each hop in the network to PC-C. First determine if PC-A can reach the ISP router Serial 0/0/1 interface with an IP address of 10.2.2.2.

C:\Users\Utraser1> **ping 10.2.2.2**

Pinging 10.2.2.2 with 32 bytes of data:

```
Reply from 10.2.2.2: bytes=32 time=41ms TTL=254
Reply from 10.2.2.2: bytes=32 time=41ms TTL=254
Reply from 10.2.2.2: bytes=32 time=41ms TTL=254
Reply from 10.2.2.2: bytes=32 time=41ms TTL=254
```

Ping statistics for 10.2.2.2:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 21ms, Average = 20ms
```

The ping was successful to the ISP router. The next hop in the network is the REMOTE router. Ping the REMOTE router Serial 0/0/1 interface with an IP address of 10.2.2.1.

C:\Users\User1> **ping 10.2.2.1**

Pinging 10.2.2.1 with 32 bytes of data:

```
Reply from 10.2.2.1: bytes=32 time=41ms TTL=253
Reply from 10.2.2.1: bytes=32 time=41ms TTL=253
Reply from 10.2.2.1: bytes=32 time=41ms TTL=253
Reply from 10.2.2.1: bytes=32 time=41ms TTL=253
```

Ping statistics for 10.2.2.1:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 40ms, Maximum = 41ms, Average = 40ms
```

PC-A can reach the REMOTE router. Based on the successful ping results from PC-A to the REMOTE router, the network connectivity issue is with 192.168.3.0/24 network. Ping the default gateway to PC-C, which is the GigabitEthernet 0/1 interface of the REMOTE router.

C:\Users\User1> **ping 192.168.3.1**

Pinging 192.168.3.1 with 32 bytes of data:

```
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
```

Ping statistics for 192.168.3.1:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

PC-A cannot reach the GigabitEthernet 0/1 interface of the REMOTE router, as displayed by the results from the **ping** command.

The S3 switch can also be pinged from PC-A to verify the location of the networking connectivity issue by typing **ping 192.168.3.11** at the command prompt. Because PC-A cannot reach GigabitEthernet 0/1 of the REMOTE router, PC-A probably cannot ping the S3 switch successfully, as indicated by the results below.

```
C:\Users\User1> ping 192.168.3.11
```

```
Pinging 192.168.3.11 with 32 bytes of data:
```

```
Reply from 192.168.1.1: Destination host unreachable.
```

```
Reply from 192.168.1.1: Destination host unreachable.
```

```
Reply from 192.168.1.1: Destination host unreachable.
```

```
Reply from 192.168.1.1: Destination host unreachable.
```

```
Ping statistics for 192.168.3.11:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

The tracert and ping results conclude that PC-A can reach the LOCAL, ISP, and REMOTE routers, but not PC-C or the S3 switch, nor the default gateway for PC-C.

- b. Use the **show** commands to examine the running configurations for the the REMOTE router.

```
REMOTE# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Embedded-Service-Engine0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	192.168.8.1	YES	manual	up	up
Serial0/0/0	unassigned	YES	unset	administratively down	down
Serial0/0/1	10.2.2.1	YES	manual	up	up

```
REMOTE# show run
```

```
<output omitted>
```

```
interface GigabitEthernet0/0
```

```
no ip address
```

```
shutdown
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 192.168.8.1 255.255.255.0
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface Serial0/0/0
```

```
no ip address
```

```
shutdown
```

```
clock rate 2000000
```

```
!
```

```
interface Serial0/0/1
```

```
ip address 10.2.2.1 255.255.255.252
```

```
<output omitted>
```

The outputs of the **show run** and **show ip interface brief** commands indicate that the GigabitEthernet 0/1 interface is up/up, but was configured with an incorrect IP address.

- c. Correct the IP address for GigabitEthernet 0/1.

```
REMOTE# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
REMOTE(config)# interface GigabitEthernet 0/1
REMOTE(config-if)# ip address 192.168.3.1 255.255.255.0
```

- d. Verify that PC-A can ping and tracer to PC-C.

```
C:\Users\User1> ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Reply from 192.168.3.3: bytes=32 time=44ms TTL=125
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125
Reply from 192.168.3.3: bytes=32 time=40ms TTL=125
Reply from 192.168.3.3: bytes=32 time=41ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 44ms, Average = 41ms
```

```
C:\Users\User1> tracert 192.168.3.3
```

```
Tracing route to PC-C [192.168.3.3]
Over a maximum of 30 hops:
```

1	<1 ms	<1 ms	<1 ms	192.168.1.1
2	24 ms	24 ms	24 ms	10.1.1.2
3	48 ms	48 ms	48 ms	10.2.2.1
4	59 ms	59 ms	59 ms	PC-C [192.168.3.3]

Trace complete.

Note: This can also be accomplished using **ping** and **tracert** commands from the CLI on the the LOCAL router and the S1 switch after verifying that there are no network connectivity issues on the 192.168.1.0/24 network.

Reflection

1. What could prevent ping or traceroute responses from reaching the originating device beside network connectivity issues?

Firewall on the PCs, access lists command, routing issues, interface is down, network delay

2. If you ping a non-existent address on the remote network, such as 192.168.3.4, what is the message displayed by the **ping** command? What does this mean? If you ping a valid host address and receive this response, what should you check?
-

Request timed out or periods (.). This means that there was no response in the default time period. Some of the items you may check: router is down, destination host is down, return route to your device and latency of the response is not more than the default time period

3. If you ping an address that does not exist in any network in your topology, such as 192.168.5.3, from a Windows-based PC, what is the message displayed by the **ping** command? What does this message indicate?

Destination host unreachable. This message indicates that there is no route to the destination as the network is not listed by the routing table.

Router Interface Summary Table

Router Interface Summary				
Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1800	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2801	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
2811	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.

Device Configs

Router LOCAL

```
LOCAL# show running-config
Building configuration...
```

```
Current configuration : 1462 bytes
```

```
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
```

```
!  
hostname LOCAL  
!  
boot-start-marker  
boot-end-marker  
!  
!  
!  
no aaa new-model  
memory-size iomem 15  
!  
!  
!  
!  
!  
!  
!  
no ip domain lookup  
ip host REMOTE 10.2.2.1 192.168.3.1  
ip host ISP 10.1.1.2 10.2.2.2  
ip host LOCAL 192.168.1.1 10.1.1.1  
ip host PC-C 192.168.3.3  
ip host PC-A 192.168.1.3  
ip host S1 192.168.1.11  
ip host S3 192.168.3.11  
ip cef  
no ipv6 cef  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface Embedded-Service-Engine0/0  
no ip address  
shutdown  
!  
interface GigabitEthernet0/0  
no ip address  
shutdown  
duplex auto  
speed auto  
!
```



```
interface GigabitEthernet0/1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.1.1.1 255.255.255.252
 clock rate 56000
!
interface Serial0/0/1
 no ip address
 shutdown
!
!
router eigrp 1
 network 10.1.1.0 0.0.0.3
 network 192.168.1.0
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
control-plane
!
!
!
line con 0
line aux 0
line 2
 no activation-character
 no exec
 transport preferred none
 transport input all
 transport output pad telnet rlogin lapb-ta mop udptn v120 ssh
 stopbits 1
line vty 0 4
 login
 transport input all
!
scheduler allocate 20000 1000
!
end
```

ROUTER ISP

```
ISP# show running-config
```

```
Building configuration...
```

```
Current configuration : 1265 bytes
```

```
!  
version 15.2  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
!  
hostname ISP  
!  
boot-start-marker  
boot-end-marker  
!  
!  
!  
no aaa new-model  
memory-size iomem 15  
!  
!  
!  
!  
!  
!  
!  
no ip domain lookup  
ip cef  
no ipv6 cef  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface Embedded-Service-Engine0/0  
no ip address  
shutdown
```

```
!  
interface GigabitEthernet0/0  
  no ip address  
  shutdown  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/1  
  no ip address  
  shutdown  
  duplex auto  
  speed auto  
!  
interface Serial0/0/0  
  ip address 10.1.1.2 255.255.255.252  
!  
interface Serial0/0/1  
  ip address 10.2.2.2 255.255.255.252  
  clock rate 56000  
!  
!  
router eigrp 1  
  network 10.1.1.0 0.0.0.3  
  network 10.2.2.0 0.0.0.3  
!  
ip forward-protocol nd  
!  
no ip http server  
no ip http secure-server  
!  
!  
!  
!  
!  
control-plane  
!  
!  
!  
line con 0  
line aux 0  
line 2  
  no activation-character  
  no exec  
  transport preferred none  
  transport input all
```

```
transport output pad telnet rlogin lapb-ta mop udptn v120 ssh
stopbits 1
line vty 0 4
login
transport input all
!
scheduler allocate 20000 1000
!
end
```

Router REMOTE

```
REMOTE# show running-config
Building configuration...
```

```
Current configuration : 1440 bytes
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname REMOTE
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
memory-size iomem 10
!
!
!
!
!
!
!
no ip domain lookup
ip cef
no ipv6 cef
multilink bundle-name authenticated
!
!
!
!
```

```
!  
!  
!  
!  
!  
!  
!  
!  
interface Embedded-Service-Engine0/0  
no ip address  
shutdown  
!  
interface GigabitEthernet0/0  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 192.168.3.1 255.255.255.0  
duplex auto  
speed auto  
!  
interface Serial0/0/0  
no ip address  
shutdown  
clock rate 2000000  
!  
interface Serial0/0/1  
ip address 10.2.2.1 255.255.255.252  
!  
!  
router eigrp 1  
network 10.2.2.0 0.0.0.3  
network 192.168.3.0  
!  
ip forward-protocol nd  
!  
no ip http server  
no ip http secure-server  
!  
!  
!  
!  
!  
control-plane
```

```
!  
!  
!  
line con 0  
line aux 0  
line 2  
  no activation-character  
  no exec  
  transport preferred none  
  transport input all  
  transport output pad telnet rlogin lapb-ta mop udptn v120 ssh  
  stopbits 1  
line vty 0 4  
  login  
  transport input all  
!  
scheduler allocate 20000 1000  
!  
end
```

Switch S1

```
S1# show running-config  
Building configuration...
```

```
Current configuration : 1565 bytes  
!  
!  
version 15.0  
no service pad  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
!  
hostname S1  
!  
boot-start-marker  
boot-end-marker  
!  
!  
no aaa new-model  
system mtu routing 1500  
!  
!  
no ip domain-lookup  
!
```

```
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
vlan internal allocation policy ascending  
!  
!  
!  
!  
!  
!  
interface FastEthernet0/1  
!  
interface FastEthernet0/2  
!  
interface FastEthernet0/3  
!  
interface FastEthernet0/4  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
!  
interface FastEthernet0/11  
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!
```

```
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 192.168.1.11 255.255.255.0
!
 ip default-gateway 192.168.1.1
 ip http server
 ip http secure-server
!
!
!
 line con 0
 line vty 5 15
!
end
```

Switch S3

```
S3# show running-config
Building configuration...
```

```
Current configuration : 1563 bytes
```

```
!
!
```



```
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname S3
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
system mtu routing 1500
!
!
no ip domain-lookup
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
!
!
!
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
```

```
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
ip address 192.168.3.11 255.255.255.0
!
```

```
ip default-gateway 192.168.3.1
ip http server
ip http secure-server
!
!
line con 0
line vty 5 15
!
end
```